

Calibration, decision time and layout area

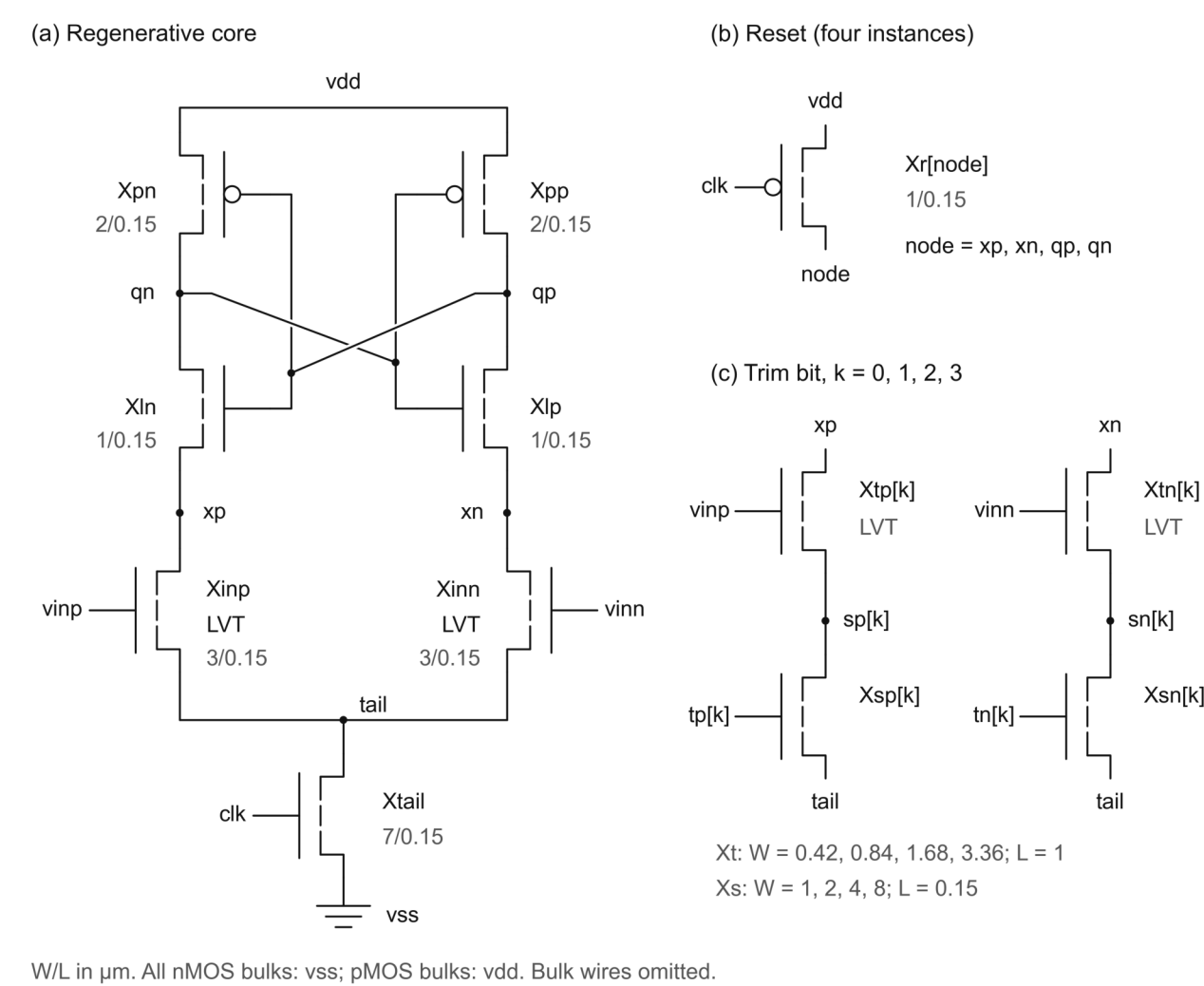
Comparator Atlas | SKY130 StrongARM characterization

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IEEE SSCS Code-a-Chip / ISSCC 2027

Circuit and decision contract

27-MOS StrongARM; LVT inputs, SVT latch.
Source-gated trim: sign + four magnitude bits.



Recorded circuit topology. Nine declared sizing candidates; the lower-energy control was evaluated after selection.

Correct, wrong or unresolved

A correct decision reaches complementary 80% / 20% output rails with the expected polarity before the deadline. Wrong polarity and unresolved outputs are counted separately.

Calibration and full-cycle core energy are evaluated from archived measurements. A later readout does not rerun SPICE or shorten the measured energy cycle.

Three sampled specifications

Calibrated SCHEMATIC: local_boundary; all 49 controlled-width-stress conditions. Both signs at finite sampled inputs through 30 mV; 1 ns deadline.

≥ 1 mV **NONE**

No feasible compared design

Mean energy / limits: not defined

≥ 3 mV **Selected: lvt_balanced_4b**

294/294 correct at all included samples

249.655 fJ mean core / cycle

≥ 30 mV **Control: lvt_base_3b**

98/98 correct at all included samples

150.531 fJ mean core / cycle

Post-hoc description of three compared designs: all-point qualification, then least mean energy. No interpolation or changed training selection.

Selected ≥ 3 mV band / same 294 samples

154.517 ps

363.217 fJ

Min. deadline - recorded decision time (sampled)

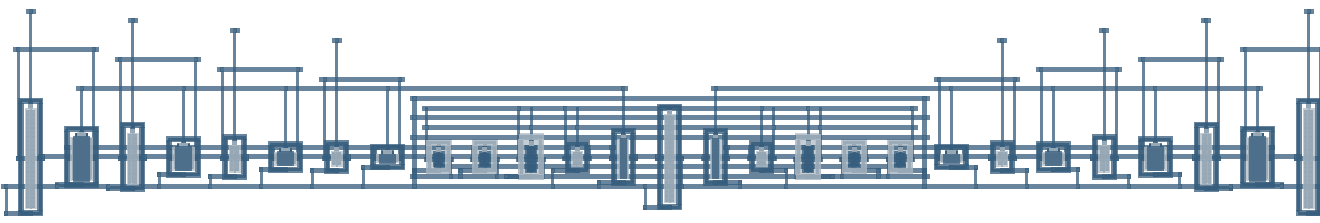
Maximum sampled core energy

20 wrong samples remain at 2 ns

Exactly the same keyed set at 1 ns and 2 ns, all at signed +/-1 mV. Locations and trim codes do not identify a physical failure cause.

Area version and archived RC

Local-pitch b1 vs adopted a1; known nonblind, code-zero: 45 PVT conditions x four signed inputs (-10, -3, +3, +10 mV). 180 keys per layout / C or RC; ngspice-47.

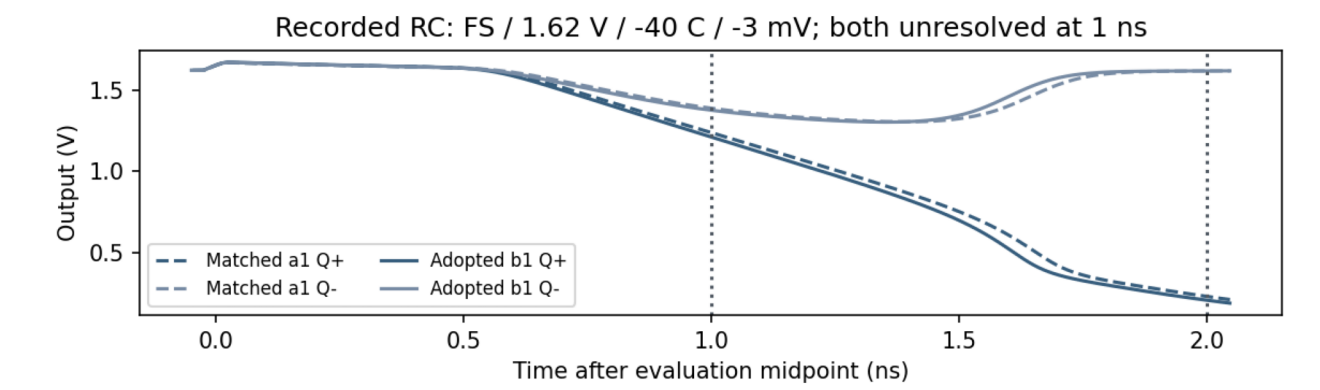


Bbox (μm^2): adopted 1873.300 / matched a1 2074.254 (9.69% less).
14 native checks: DRC 0 / LVS / negatives; not signoff.

Correct / 180	1 ns	2 ns
Both C	170/180	180/180
a1 RC	156/180	180/180
b1 RC	158/180	180/180

RC mean: 421.04 -> 412.88 fJ / cycle

1 ns RC unresolved: a1 24 -> b1 22; zero wrong.
Only SS / 1.62 V / 125 C / +/-10 mV improve; FS remains late.
2 ns primary, prospectively declared; no lost correct keys.



Matched a1 -> b1; worst RC: 1.811 -> 1.760 ns.
~1.94% energy change is not robust global PPA.

Archived RC-deck outcomes; model physical fidelity not yet qualified.

No qualified PEX or silicon prediction. C-only is not independent ground truth; RC-versus-C differences do not isolate resistance.

Finite samples, not foundry Monte Carlo/yield, noise/jitter/PVT confidence, timing signoff or a worst-cycle guarantee. Core excludes drivers and calibration/controller infrastructure.

References: Razavi (2015), DOI 10.1109/MSSC.2015.2418155; Li, Xu & Iizuka (2022), DOI 10.1007/s10470-022-01992-6.

Original code: MIT; SKY130: Apache-2.0. GitHub Copilot assisted implementation, experiment automation, figures and documentation; the author is responsible.

[Notebook](#)

[Colab](#)